1 Edge Detection

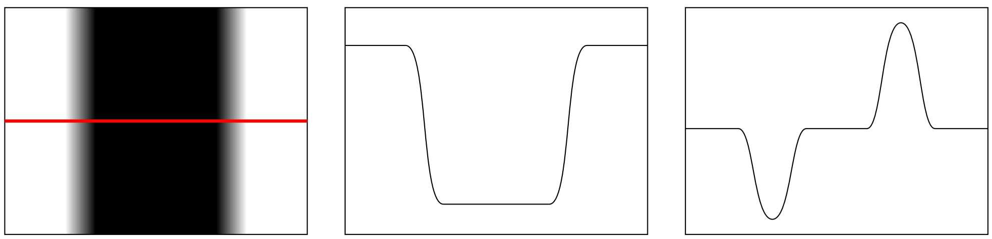
In image processing, an edge represents a discontinuity or rapid transition in the image intensity function, $I(x, y)$. Mathematically, edges are defined as the locus of points where the gradient magnitude, $\|\nabla I(x, y)\|$, is locally maximized.

1.1 1D Analysis via Scanlines

Consider a 1D intensity profile $f(x) = I(x, y_0)$ extracted along a horizontal scanline.

- **Intensity Function $f(x)$:** An edge is characterized by a steep step or ramp.
- **First Derivative $f'(x)$:** The spatial derivative transforms intensity ramps into distinct local extrema.

Core Principle: Image edges correspond to the local extrema (maxima or minima) of $\frac{df}{dx}$.



1.2 The Effects of Noise

Real-world images are corrupted by high-frequency sensor noise $\eta(x)$, yielding an observed signal $\tilde{f}(x) = f(x) + \eta(x)$.

1.2.1 Noise Amplification

Because differentiation acts as a high-pass filter, it heavily amplifies the noise component. The derivative of the noise, $\frac{d\eta}{dx}$, often eclipses the true signal derivative, making direct extrema extraction ill-posed.

1.2.2 Smoothing

To regularize the problem, $\tilde{f}(x)$ is pre-conditioned via convolution with a low-pass smoothing kernel $g(x)$, typically a Gaussian $g(x, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right)$. Edge locations are then identified by:

$$x_{\text{edge}} = \arg \operatorname*{local\,max}_x \left| \frac{d}{dx}(\tilde{f} * g) \right|$$

Theorem 1.1 (Derivative Theorem of Convolution): *Because differentiation is linear and shift-invariant, it commutes with the convolution operator,*

$$\frac{d}{dx}(f * g) = f * \frac{dg}{dx}.$$

Computational Advantage: *We can pre-compute the derivative of the smoothing kernel,* $\frac{dg}{dx}$*, and convolve the image directly with this filter. This reduces a sequential two-step process (smoothing, then differentiating) into a single linear filtering operation.*

1.3 Tradeoff: Smoothing vs. Localization

The choice of the Gaussian standard deviation, σ, governs the scale of edge detection and presents a fundamental uncertainty principle:

- **Small σ (Fine Scale):** Yields high spatial localization and preserves fine detail, but provides poor noise attenuation, resulting in a high false-positive rate.
- **Large σ (Coarse Scale):** Provides strong noise suppression and extracts dominant macro-structures, but causes poor spatial localization (edge displacement) and blurring of fine details.

1.4 Designing an Edge Detector

To formalize performance, a robust differential edge detector (such as the Canny edge detector) optimizes three specific criteria:

1. **Good Detection (High SNR):** Maximize the probability of detecting true edges while minimizing false positives triggered by noise artifacts.
2. **Good Localization:** Minimize the variance of the edge position estimate (detected pixels must be as close as possible to the true physical edge).
3. **Single Response:** Ensure the detector produces exactly one localized pixel for each true edge, actively suppressing thick, multi-pixel responses (e.g., via non-maximum suppression).

1.4.1 Advanced Cues

Beyond scalar intensity gradients, modern computer vision systems integrate diverse cues to robustly identify boundaries:

- Multimodal gradients (e.g., abrupt changes in color vectors or texture statistics).
- Gestalt priors such as continuity, collinearity, and closure.
- Semantic, top-down priors (e.g., matching edge fragments to expected shapes of known objects).

1.5 Canny Edge Detector

`[Canny Edge Detector]`

2 Blob Detection

In computer vision, a **blob** is defined as a contiguous region of pixels exhibiting spatial coherence in properties such as intensity, color, or texture, distinct from the surrounding background. Blob detection algorithms aim to localize these distinct structures, which often correspond to objects or object parts in a scene.

2.1 The Laplacian Operator

To detect regions bounded by rapid intensity transitions, we analyze the second spatial derivative of the image. The isotropic 2D analog of the second derivative is the **Laplacian operator**.

For a continuous 2D image intensity function $f(x, y)$, the Laplacian, $\nabla^2 f$, is defined as the trace of the Hessian matrix (the sum of the unmixed second-order partial derivatives):

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

2.2 Laplacian of Gaussian (LoG)

Because second-derivative operators severely amplify high-frequency spatial noise, direct application to raw images is ill-posed. The signal must first be regularized via convolution with a Gaussian low-pass filter, $G_\sigma(x, y)$.

By the associative property of convolution, we can pre-compute the **Laplacian of Gaussian (LoG)** filter and convolve it directly with the image: $\nabla^2(f * G_\sigma) = f * \nabla^2 G_\sigma$.

Defining the 2D Gaussian with variance σ^2:

$$G_\sigma(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

Applying the Laplacian operator analytically yields the LoG filter:

$$\nabla^2 G_\sigma(x, y) = \frac{x^2 + y^2 - 2\sigma^2}{\sigma^4} G_\sigma(x, y)$$

The 3D morphology of the LoG operator resembles an inverted ”Mexican hat.” It is characterized by a prominent central peak surrounded by an isotropic negative annulus that asymptotically approaches zero.

2.3 Discrete Approximation of the LoG

For digital images, the continuous Laplacian is approximated using finite differences. The second-order partial derivatives are modeled in 1D as:

$$\frac{\partial^2 f}{\partial x^2} \approx f(x+1, y) - 2f(x, y) + f(x-1, y)$$
$$\frac{\partial^2 f}{\partial y^2} \approx f(x, y+1) - 2f(x, y) + f(x, y-1)$$

These linear operations correspond to convolution with the following 1D spatial kernels:

- **x-kernel:** $\begin{bmatrix} 1 & -2 & 1 \end{bmatrix}$
- **y-kernel:** $\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$

Superimposing these orthogonal kernels yields the standard 4-connected discrete Laplacian approximation:

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & -2 & 1 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 \\ 0 & -2 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

2.4 Blob Detection Strategy

Detecting blobs via the LoG operator relies on finding the spatial extrema of the filter response:

- **Convolution & Extrema:** The image is convolved with the LoG kernel. The spatial coordinates (x, y) of local extrema in the response correspond to blob centers.
- **Template Matching:** The LoG kernel inherently acts as a matched filter. It yields a maximal response when its central region aligns with a blob of corresponding scale (radius $r \approx \sqrt{2}\sigma$) and contrasting background.
- **Polarity Independence:** Dark blobs on bright backgrounds produce strong positive extrema, while bright blobs on dark backgrounds yield strong negative extrema. Both designate valid blob detections.

3 Corner Detection

3.1 Harris Corner Detector